

Can MLLMs reason in topological space?

Reasoning Task

Question:

In which option can the metal ring be removed from the white ring?



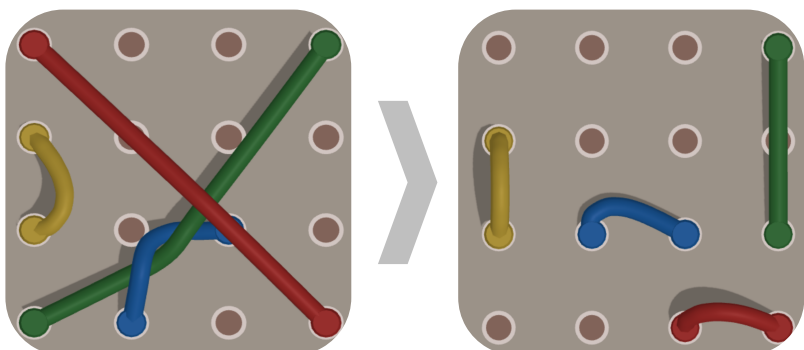
A

✓ B

Planning Task

Plan actions:

Rearrange the ropes in the given image so that no two ropes cross each other.



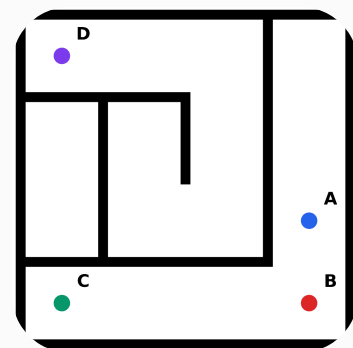
Initial State

Final State



Evaluating topological intuition across five categories

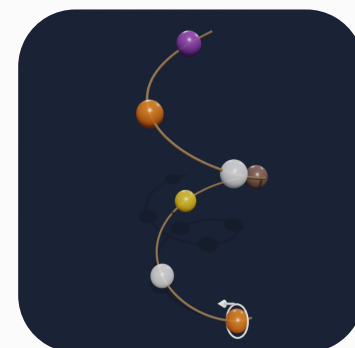
Reasoning



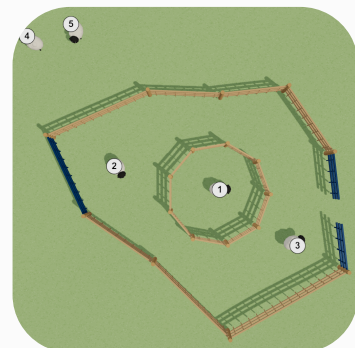
Maze



Assembly



Bead



Sheep



Knot

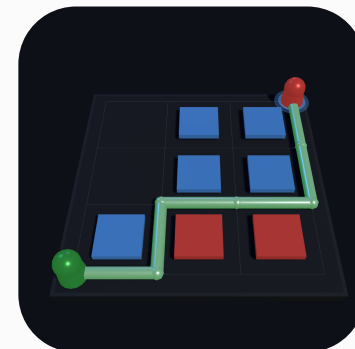


MindTopo

Planning



Pipe



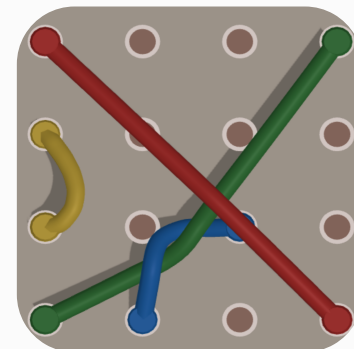
One Stroke



Swap



Chat Noir



Untangle

Continuity

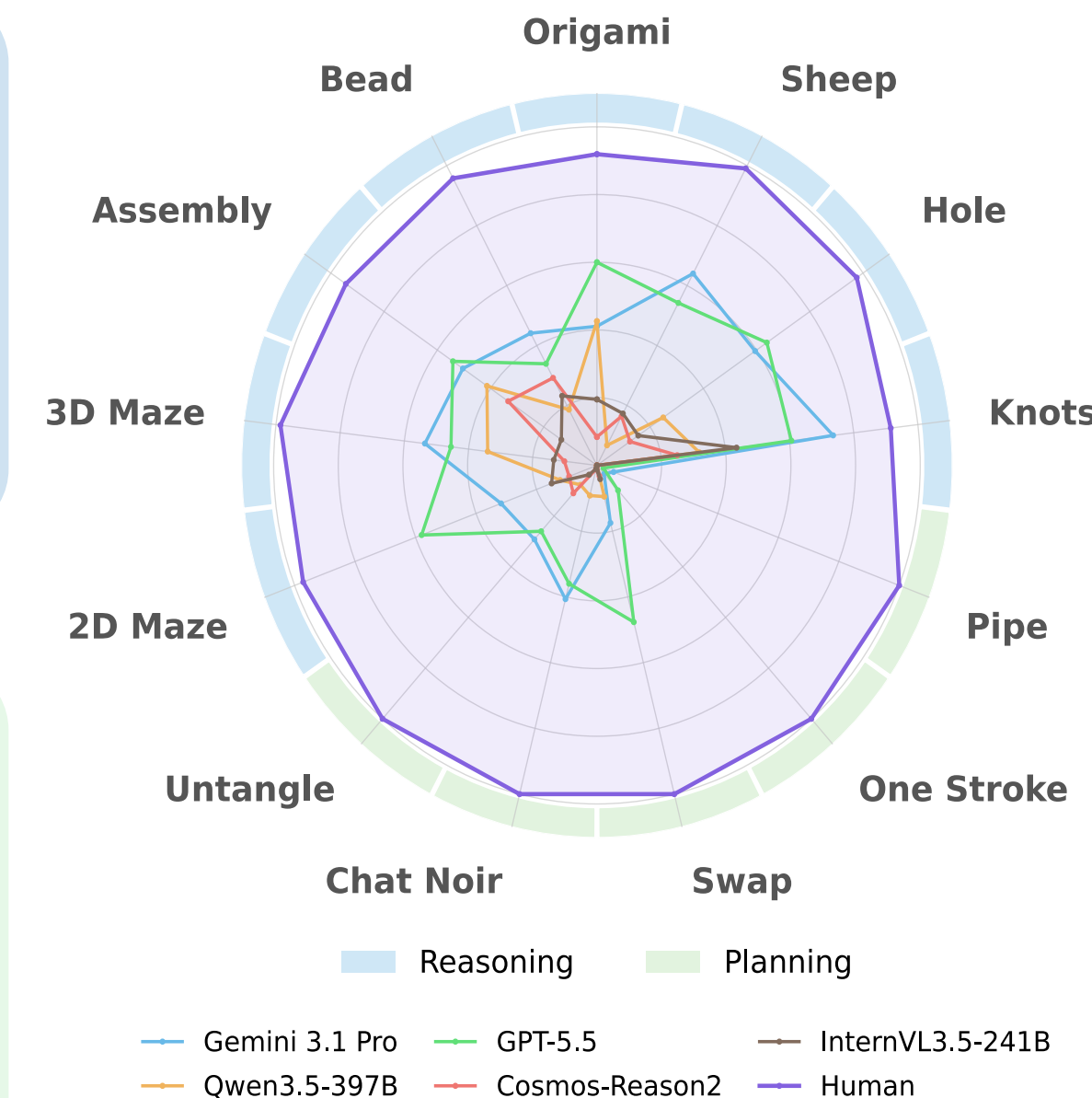
Separation

Order

Enclosure

Knots

Current models still struggle



Topological spatial reasoning remains a **blind spot**